

STA 5635 HW 8

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Part a

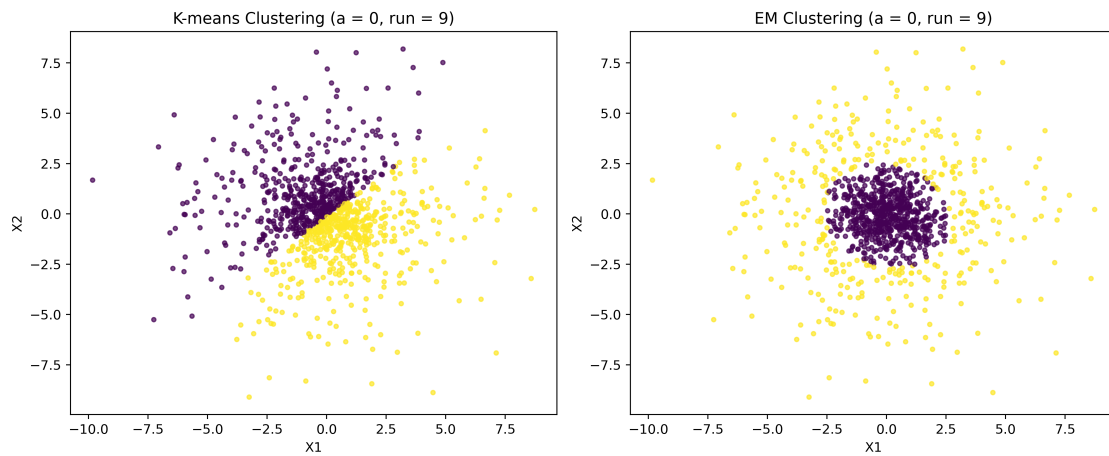


Figure 1: Clustering Results for K-means and EM, $a=0$

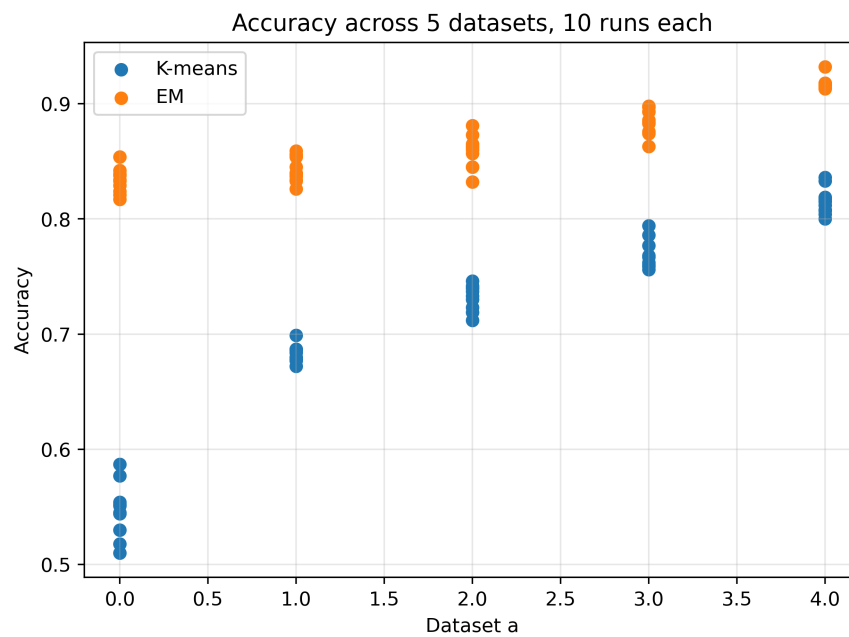


Figure 2: Accuracy for each run vs. dataset parameter a

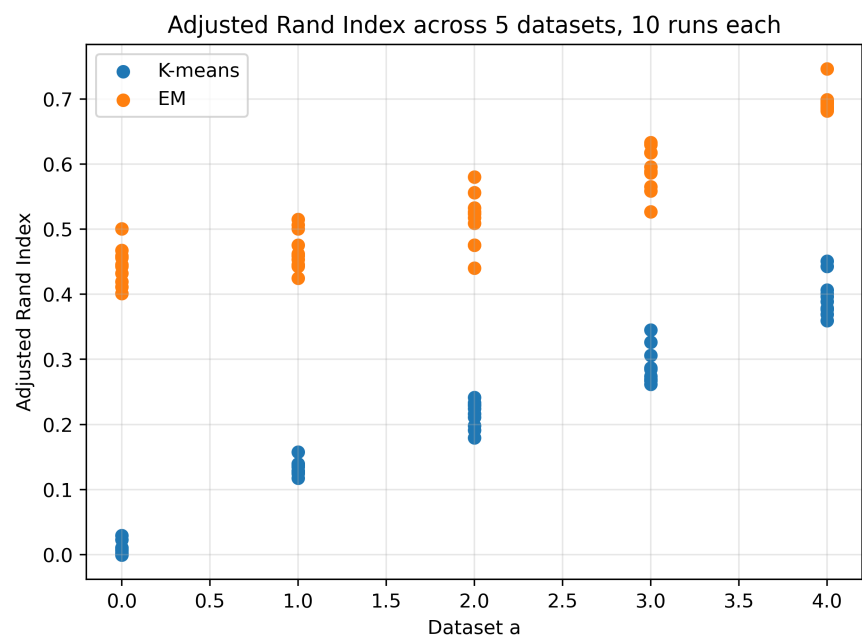


Figure 3: Adjusted Rand Index for each run vs. dataset parameter a

Part b

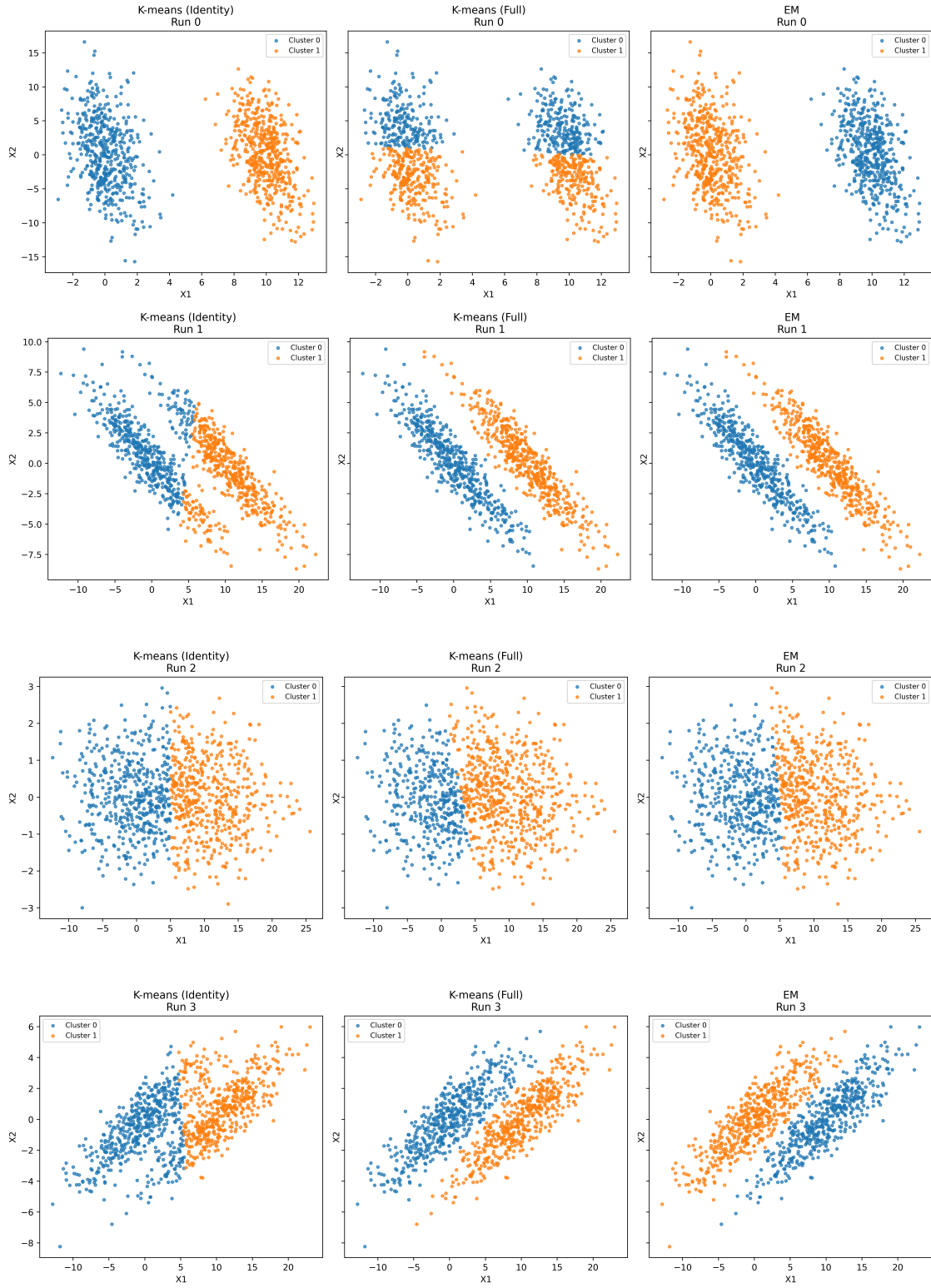


Figure 4: Clustering results for the first four runs.

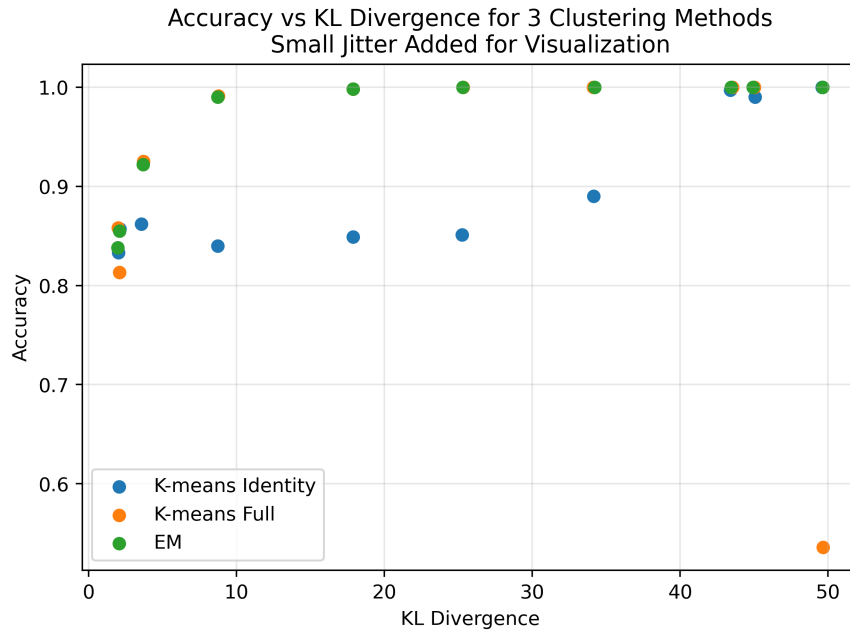


Figure 5: Accuracy vs. KL Divergence for the 3 methods on the 10 runs. Small jitter added to x values for visualization.

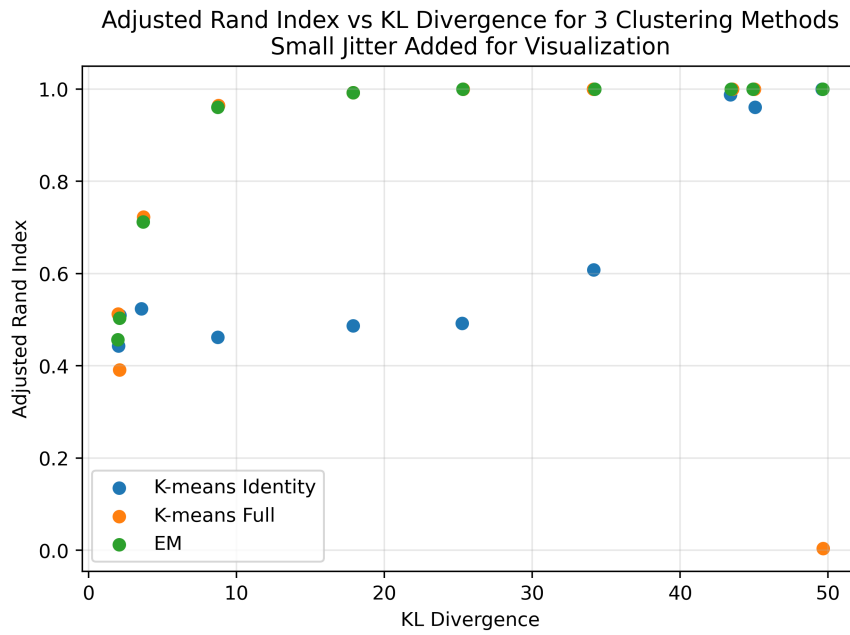


Figure 6: Adjusted Rand Index vs. KL Divergence for the 3 methods on the 10 runs. Small jitter added to x values for visualization.

Run	KL	Acc KM (Id)	ARI KM (Id)	Acc KM (Full)	ARI KM (Full)	Acc EM	ARI EM
0	49.662	1.000	1.000	0.536	0.004	1.000	1.000
1	17.821	0.849	0.487	0.998	0.992	0.998	0.992
2	2.026	0.833	0.443	0.813	0.391	0.838	0.456
3	8.716	0.840	0.462	0.991	0.964	0.990	0.960
4	2.000	0.857	0.509	0.858	0.512	0.855	0.504
5	25.273	0.851	0.492	1.000	1.000	1.000	1.000
6	44.941	0.990	0.960	1.000	1.000	1.000	1.000
7	34.212	0.890	0.608	1.000	1.000	1.000	1.000
8	43.471	0.997	0.988	1.000	1.000	1.000	1.000
9	3.636	0.862	0.524	0.925	0.722	0.922	0.712

Table 1: KL Divergence, Accuracy, and Adjusted Rand Index for each method. Each row is the result from 1 run.